

Visions of sustainable finance. Media Coverage of Capital Markets and Food Systems

Abstract

Capital markets are positioned as essential to sustainable food system transitions, yet how this consensus is constructed and reproduced in public discourse remains poorly understood. This paper examines how global news media frame the relationship between capital markets and sustainable transitions in food systems. Using a dataset of 3,200 news articles (2000–2025) from major international outlets, we employ two computational text analysis methods - Named Entity Recognition (NER) and Structural Topic Modeling (STM) to analyze the main actors and themes in sustainable finance discourse. Results show that coverage is dominated by international financial institutions and multinational corporations, framing sustainability as a technocratic and market-driven project not necessarily focused on more holistic interpretations. Additionally, we examine the role of media in shaping public opinion, influencing policy, and framing key societal issues. Specifically, we test how political bias affects the discourse on sustainable food systems. We find that while left-leaning outlets focus on food security, smallholders, and sustainable development, right-leaning outlets place greater emphasis on market-oriented themes such as profitability, ESG investing, and supply shocks. Our findings suggest a tension at the heart of sustainable investment in food systems. The idea that capital markets' have "disruptive" potential is largely discursively constructed through media narratives that accept financial authority as given in sustainability governance rather than challenge existing structures.

Keywords: sustainable finance; food systems; media framing; structural topic modelling; named entity recognition; political bias

1. Introduction

Food systems are at the center of the climate crisis. They are among the sectors most vulnerable to climate change, while simultaneously accounting for a third of global greenhouse gas emissions (Crippa et al., 2021; Hultgren et al., 2025). Addressing these two urgent challenges has generated calls for a sustainable transformation of agrifood systems that reduces emissions and improves resilience.

Delivering on such a transformation requires large-scale investment. Recent estimates suggest that meeting the commitments of the Sustainable Development Goals (SDGs) and the Paris Agreement would require an annual investment equivalent to roughly 2.2 per cent of global GDP, an amount far beyond what public finance can provide (GFANZ, 2023). As a result, governments and international organizations such as multilateral development banks and United Nations agencies recognize that private capital and market-based financing are vital components of climate financing (Blended Finance Taskforce, 2021; UNEP, 2023; Finance for Food Systems Transformation, 2023).

However, while multilateral institutions increasingly position capital markets as vital for sustainable food system transformations, the legitimacy of this approach partly depends on the media's power to shape the public narratives through which sustainability challenges are defined and capital market-based solutions are accepted. Media coverage is central to how the governance and financing of food systems are understood because it shapes which issues gain attention and how they are evaluated. Media outlets act as interpretive institutions that translate complex and contested policy debates into publicly accessible narratives that can be used to shape how risk is framed, and influence investor behavior.

While there is a well-established literature on media framing of climate change and environmental policy (Boykoff & Boykoff, 2004; Carvalho, 2002; Chinn et al., 2020; Schmidt et al., 2013), and a rapidly growing body of work on the increased influence of capital markets in agrifood systems (Clapp, 2019; Clapp & Isakson, 2018; Fairbairn, 2014), the intersection of these literatures remains unexplored. As a result, we still lack a systematic understanding of how media coverage frames the legitimacy of capital markets in sustainable food system transitions and why this framing matters. Addressing this gap is important for two reasons. First, because sustainability transitions can be understood as sociotechnical imaginaries or shared, socially embedded understandings of how the world works and what a desirable future should look like, the media plays a constitutive role in defining what these transitions mean. Given the plurality of transition pathways, what is perceived as the 'right' transition remains contested - frequently between technocentric (market based) and ecocentric approaches (foregrounding justice and ecological limits). Thus the way in which media narratives select and highlight certain problems, definitions, actors, and instruments while marginalizing others, directly shapes which version of sustainability becomes dominant (Entman, 1993; McCombs & Shaw, 1972; de Vreese, 2005). Second, media framing also shapes what becomes politically feasible by influencing

how both publics and decision-makers interpret complex policy issues. (Barnes and Hicks, 2018; Besley and Burgess, 2002; Bromley-Trujillo and Karch, 2021; Ferraz and Finan, 2008; Strömberg, 2001). Together, media narratives can both shape our understanding of the problem, but also constrain the range of responses that are perceived as legitimate.

In this paper, we address this gap by examining how global news media have constructed the relationship between capital markets, agrifood systems, and sustainability. Our analysis is guided by three research questions. First, which actors and organizations dominate media coverage at this intersection. Second, what are the dominant themes through which finance and sustainable agriculture are linked in global news coverage, and how have these themes evolved over time? Third, how do political orientation and institutional characteristics of media outlets shape the framing of capital markets as solutions to sustainability challenges in food systems? To address these questions, we analyze a corpus of over 3,000 major news publications using a mixed computational text analysis approach. Specifically, we employ Named Entity Recognition (Grimmer et al., 2022) to map the actors structuring the discourse and Structural Topic Modeling (Roberts et al., 2014) to identify latent themes and systematically test how their prevalence varies by political bias and outlet type. This approach allows us to move beyond anecdotal or case-based analyses and to identify structural patterns in how finance-led agrifood transitions are publicly constructed.

2.Theoretical Framework

A large body of scholarship demonstrates that the media influences political processes by shaping issue salience, defining problems, and legitimizing particular actors and policy responses (Barnes and Hicks, 2018; Kingdon, 1995; McCombs, 1997). Importantly, this literature does not suggest that the media determines policy outcomes directly or uniformly, but that media narratives condition the interpretive environment within which political and economic decisions are made (McNair, 2014). As such, identifying which actors and topics dominate media discussions of finance and sustainability is essential.

Here, we draw on two closely related strands of scholarship, agenda-setting theory and framing theory, to conceptualize this influence. Agenda-setting theory emphasizes the media's capacity to shape which issues are perceived as important by allocating attention across topics (McCombs & Shaw, 1972). Consequently, media coverage influences what audiences think about rather than what to think. Framing theory, on the other hand, focuses on how issues are presented once they become salient. As Entman (1993) argues, frames select certain aspects of reality and make them more prominent in order to promote specific problem definitions, causal interpretations, and policy evaluations. Together, these theories highlight that media influence operates through structuring attention and interpretation, rather than by directly determining preferences or behavior.

A large body of empirical evidence demonstrates media can shape political behavior and institutional responses across a range of policy domains (Strömberg, 1999; Barnes and Hicks, 2018; Ferraz and Finan, 2008; Besley and Burgess, 2002; Bolstad and Victor, 2024; Bromley-Trujillo and Karch, 2021). Our purpose is not to argue that media effects exist but rather to draw on this literature as a theoretical basis for examining how media influence operates in contexts characterized by high complexity, uncertainty, and contested expertise - conditions that characterize debates over sustainable finance and agrifood system transformation.

Existing studies in environmental communication have examined the ideological polarization of climate change coverage (Chinn et al., 2020; McCright & Dunlap, 2014; Schmidt, Ivanova, and Schäfer, 2013), and a smaller literature has explored media framing of sustainability within the food system (Rust et al., 2021; Simmonds et al. 2025). While these studies find that media discourse on sustainable agricultural practices is highly polarized, they focus on sector-specific practices and debates over the assignment of environmental responsibility, rather than on the role of capital markets in shaping sustainable transitions.

In sum, these studies suggest that variations in how the media cover the role of capital markets in sustainable agrifood systems can have significant implications for how transitions are conceived and pursued; however, research on this intersection is limited.

2.2 Financing sustainable agrifood systems.

Financial investment in the agrifood sector has increased significantly since the 2009 financial crisis driven by expectations of rising global food demand and narratives positioning agriculture as a resilient investment class (Clapp, 2019). This expansion of financial investment has an ambivalent relationship with sustainability. On the one hand, it has the potential to unlock the transformative amount of available financial capital to fund sustainable agricultural practices, given finance's role as a potential leverage point for sustainability when directed towards long-term regenerative investment (Mazzucato & Semieniuk, 2018; Stephens, 2021, Robinson et al, 2024). Sustainable finance instruments such as ESG rated equities, green bonds, and sustainability-linked loans are frequently presented as mechanisms through which financial returns can be aligned with environmental performance and social responsibility.

At the same time, political economy scholarship emphasizes that finance is not a neutral tool for sustainability. The types of financial tools available and the mentality of investors can diminish the resilience of food systems and their overall capacity for regeneration (Stephens and Clapp, 2021). In this sense, recent work on the role of finance in agrifood systems has highlighted that speculative investment can contribute to food insecurity (Ghosh, 2010; Martin & Clapp, 2015; Isakson et al., 2023) and environmental degradation (Fairbairn, 2014; Clapp & Isakson, 2018). Thus, whether finance plays a transformative or detrimental role depends on how financial and policy actors understand sustainability and the particular menu of interventions that this understanding creates.

Theoretically we argue that the media plays a key role in this process through two complementary mechanisms. First, investors rely on heuristic measures to transform uncertainties into assessable risk (Brooks et al 2015, Gray 2013; Mosley 2003). Media coverage contributes to this process by influencing which sustainability issues become salient and how they are interpreted. In doing so, the media operates as one node within a wider knowledge infrastructure (alongside analysts, rating agencies, NGOs, and policy institutions) that helps to transform complex ecological uncertainties into quantifiable risk.

Secondly, the media also plays an important role by legitimizing particular actors and policy responses (Barnes and Hicks, 2018; Kingdon, 1995; McCombs, 1997). Under conditions of uncertainty, financial and policy actors rely on socially visible signals to evaluate which actors, instruments, and solutions are credible and appropriate (Deephouse and Suchman, 2008; Pollock and Rindova, 2003). Media coverage can contribute to this process by highlighting certain organizations (such as multilateral development banks, financial institutions, and large corporations) and specific instruments such as ESG investing or green bonds. This framing and setting from the media helps normalize these actors and approaches as authoritative and viable pathways for addressing sustainability challenges.

By focusing on certain instruments and practices, the media can contribute to the construction and diffusion of dominant interpretations. For example, narratives centered on ESG investing or supply-chain disclosure can help paint capital markets as force for good, but at the same time can sideline alternative approaches such as agroecology, alternative food networks, or social finance initiatives rooted in local food sovereignty (Stephens, 2021). Conversely, narratives that emphasize the harms associated with speculation, land grabs or corporate consolidation may be sceptical of the capacity of capital markets to deliver just transitions.

Conversely, actors and approaches that receive limited media attention, such as smallholder organizations, agroecological practices, or alternative food networks, are less likely to be perceived as central to the transition. On the flipside, by emphasizing certain actors the media can help us identify pivotal leverage points for redirecting sustainability trajectories. As research on sustainability transitions increasingly emphasizes, influence over transition pathways is not evenly distributed across actors, but instead concentrated among a relatively small set of organizations that occupy structurally central positions within governance and financial systems. As such, interventions targeting central actors can have system-wide ripple effects (Dordi et al. 2022).

Here, we extend this insight to media discourse on sustainable agrifood systems. We argue that, given the media functions as interpretive institutions, actors and instruments who are most visible in media coverage are more likely to shape how sustainability challenges are defined and which solutions are rendered legitimate.

Thus, this scholarship shows that the media not only communicates environmental information but also constructs narratives that allocate responsibility, define problems, and

shape political responses. This is crucial given that, as theorized, media framing entails that the dominant discourses around the role of capital markets in the food system will have a profound impact on the direction of its sustainable transition.

2.3 Political Bias and Divergent Narratives of Sustainability

In the previous section we argued that the media has both agenda setting power with regards to the content that they choose to publish but also by how they frame those topics. While media influence is general, this influence is not uniform across outlets. News organisations differ in how they convey content (Earle and Hodson, 2022). Here we explain what our expectations of differences in narrative salience are based on the political orientation of particular media sources.

First, research documents that left- and right-leaning media portray policy issues in distinct ways and that the rise of partisan media has contributed to political polarization and differential support for policies with conservative and liberal actors advancing competing narratives about responsibility, regulation, and the role of markets (Bernhardt et al., 2008; Earle & Hodson, 2022; Prior, 2013). This dynamic is particularly visible in sustainability debates, where belief in climate change has become a litmus test of political ideology and partisan actors mobilize competing narratives (McCright et al., 2014; Chinn et al., 2020). The ideological bias of an outlet is therefore likely to influence which aspects of sustainable finance receive attention and how they are framed.

Theoretically, differences in ideological framing arise from both supply-side dynamics, where the preferences of the agents who own and work for the news organization shape the content and themes, and demand-side explanations where media outlets want to maximize audiences and so tailor their content to attract certain audiences (Prat and Strömborg, 2013). Consequently, media outlets translate sustainability debates into ideologically familiar frames that allow readers to make sense of the issue quickly and consistently with prior beliefs. In equilibrium, a liberal individual will get more relevant information from a liberal media source, and a conservative individual will get more relevant information from a conservative outlet (Chen and Suen, 2023⁰⁸). Thus, we expect that news organizations' political orientations will structure their portrayals of finance and sustainability in distinct, ideologically consistent ways where conservative outlets, by definition, defend the status quo, while liberal outlets challenge it.

Applying this logic to the case of sustainable food systems, we anticipate systematic differences in how left- and right-leaning outlets frame the role of finance. We expect that right leaning outlets will adopt a market-liberal understanding of sustainability where sustainability problems are framed as problems of information asymmetry and capital allocation and profitability. Thus, they will emphasize themes that position finance as a positive tool of greening the food system in line with the win-win logic common to ecological modernization (Bäckstrand & Lövbrand, 2006; Carter, 2018). As such, we expect these

outlets to emphasize themes such as corporate sustainability strategies, ESG investing (both for and against), and profitability-oriented narratives that portray capital markets as the primary vehicle for greening food systems.

In contrast, we expect left-leaning outlets to foreground the concerns raised by critical scholarship on finance and food systems, such the role of capital in propagating food insecurity (Ghosh, 2010), power asymmetries (Clapp, 2019), vulnerabilities of smallholders (Isakson et al., 2023), and the structural drivers of ecological degradation (Fairbairn, 2014). As such, we expect these outlets to highlight more holistic issues including food security, farmer livelihoods, smallholder perspectives and development. From these expectations follow our first set of hypotheses.

H1a. Conservative outlets will place greater emphasis on market-based themes in their coverage of sustainable food systems.

H1b. Liberal outlets will place more emphasis on agroecological themes such as smallholders, and poverty alleviation.

2.4 Issue Ownership and Financial Media

Political orientation is not the only predictor of what topics will be emphasized by media outlets. Media sources are also selective about the types of issues they cover (Brutger & Strezhnev, 2022) and specialized outlets exercise agenda-setting power over what counts as the key problems and solutions within their domain.

This selectivity is shaped by what has been termed “issue ownership” where specialized news outlets often establish interpretive authority over specific issue domains, leading to divergence in topics covered and how they are framed (Wolton, 2019). In the context of the sustainability of agrifood systems, previous work suggests that issue ownership shapes how sustainability practices are portrayed. Simmonds et al. (2020), for instance, show that when sustainable agricultural practices are covered by general interest media sources they are framed in terms of environmental responsibility and public policy, whereas farming-sector media tend to emphasize threats to farmer livelihoods and economic viability. These findings show how specialized outlets, including financial news outlets, can frame sustainability debates in ways that speak directly to their core audiences.

Financial news outlets such as the *Financial Times*, *The Wall Street Journal*, and *Bloomberg*, have institutionalized authority over investment practices, risk management, and market performance. On the supply side, their professional routines, expert sources, and editorial priorities mean they are more likely to foreground issues that align politically or ideologically with financial actors. On the demand side, readership is concentrated in financial and corporate sectors, creating strong incentives to focus on information relevant to investment decision-making, including ESG metrics, risk mitigation, and strategies for sustainable profit growth (Clapp, 2019; GSIA, 2018; Robinson et al., 2022). We therefore expect financial news outlets to emphasize themes central to investment decision-making,

positioning capital markets as the central actors in this transition. This leads to our second set of hypotheses:

H2a. Finance-specialized outlets will emphasize investment themes such as market performance and ESG metrics.

H2b. General-interest outlets will emphasize holistic and multifunctional themes such as on food security and governance.

In summary, H1a and H1b address how political orientation shapes coverage, while H2a and H2b address how outlet specialization shapes coverage. Taken together, these insights highlight how media power, political bias, and issue ownership interact to shape which visions of agrifood transitions gain prominence and which are sidelined.

To address these dynamics empirically, we rely on Named Entity Recognition (NER) to identify the key actors and organizations that dominate the discourse (RQ1), and Structural Topic Modeling (STM) to infer the themes in the discourse (RQ2) and to test whether political orientation (RQ3) and outlet type (RQ4) systematically shape topic prevalence.

3. Research Design (1500-2000)

To map media discourse at the intersection of food systems, finance, and sustainability, we rely on a combination of named entity recognition and structured topic modelling analysis of news articles in order to answer our research questions.

3.1. Data Collection and Processing.

Our main source of data is news articles exported from the *Nexis Uni* Database, a repository of news from over 10,000 news, legal, and business sources. The aim was to capture the intersection of news reports on finance, agrifood systems, and sustainability over the last 25 years. We designed a Boolean query to identify articles that explicitly link financial themes to agrifood and sustainability ones. The query¹ searched for the co-occurrence of at least one keyword from each of the three conceptual clusters in article headlines as shown in Table 1. Restricting the search to headlines and lead paragraphs ensures we capture discursive salience, filtering for articles that actively focus on this intersection rather than mentioning it tangentially.

¹ Query: (hlead (("sharehold*" OR "investor*" OR "investment*" OR "capital market*" OR "finan*" OR "stock market*" OR "listed compan*" OR "stock ownership"))) AND hlead (("food*" OR "agri*" OR "agro*" OR "farm*")) AND hlead (sustaina*)English

Table 1.

Cluster	Keywords
Finance Cluster	“sharehold*”, “investor*”, “investment*”, “capital market*”, “finan*”, “stock market*”, “listed compan*”, “stock ownership”
Agrifood Cluster	“food*”, “agri*”, “agro*”, “farm*”
Sustainability Cluster	“sustaina*”

In addition to this, we restricted the search to *Major World Publications* published in English. We also restricted the temporal window from January 2000 to August 2025. This 25 year span also allows us to analyze changes in the narrative before and after the adoption of the SDGs in 2015, a key global policy turning point. The final sample exported 3,631 articles.

After extracting our database, we follow (Gruber, 2020; Ordonez-Ponce et al., 2024) to exclude duplicate articles by identifying highly similar texts. Specifically, we use a combination of word similarity (or the Levenshtein distance) to measure the difference between two sequences of text (Levenshtein 1966), and the relative distance comparing two texts by the words they contain, not by their character sequence. This allows us to exclude articles in instances where texts are similar but not identical. Following (Gruber, 2020), we apply a similarity threshold of 0.97 and a relative distance threshold of 0.2. Finally, we manually excluded from the sample non-news sources such as trade reports, analytical briefings, and peer-reviewed journal articles to ensure that the corpus reflected journalistic discourse rather than technical and/or academic writing. Following these steps, the final sample consisted of 3,189 news articles.

Following duplicate elimination, each article’s metadata was cleaned and standardized to include publication title, country of origin, and publication date to ensure that each source or country was represented by a single standardized label (for example, *United Kingdom* and *UK* were merged into *UK*, and *New York Times* and *NYT* into *New York Times*), and had the same date format (dd/mm/yyyy). In addition to the standard metadata, we also included additional metadata to test our hypothesis/RQs. First, to test the influence of political biases in news reporting, we extracted the political orientation of each publication. This was done either by reference to explicit editorial stances or by referencing third-party datasets such as *Media Bias/Fact Check* and *AllSides*. We grouped outlets as either *Left*, *Center*, or *Right*. Where no explicit bias rating was available (for instance, in specialized trade news sources) we conservatively coded them as *Center*. To test whether discourse surrounding financing of agrifood systems differs in finance specific outlets from generalist ones, we included a dummy variable where 1 indicates a finance specific outlet (i.e. *The Financial Times*) and 0 a generalist outlet (i.e. *The Guardian*). Finally, we also included a dummy variable to reflect whether an article was published before or after the adoption of the SDGs in September

2015. This step enriched our metadata and allowed us to test how institutional, political, and temporal dynamics shape the framing of finance and sustainable food systems.

3.2. Empirical Approach

To answer our research questions, our empirical approach uses a combination of Named Entity Recognition (NER) as well as Structural Topic Modeling (STM) to examine the themes and actors discussed in media discussions of sustainable finance and food systems as well as to explain how these themes vary across metadata. These methods help us capture both *who* dominates the conversations and *what* institutional dynamics structure it.

3.2.1 Named Entity Recognition

To address RQ1 and answer which organizations/actors dominate the conversation, and what this implies about whose vision of “sustainable transitions” gets amplified, we leverage Named Entity Recognition on our corpus.

In order to identify the most common organizations in the corpus, we rely on Named Entity Recognition (NER). This helps in understanding the context and focus of the articles and importantly, identifies if there are any dominant actors in the discourse around sustainable finance and food systems. A named entity is a real object that you can refer to by a proper name. It can be a person, a location, an organization, or any other entity such as products. These are important because they reveal the place or organization the text is talking about (Benoit & Matsuo, 2018; Honnibal & Montani, 2017, Vasiliev, 2020 p.29).

To examine the relational structure of actors in the discourse, we construct a co-occurrence network based on named entities identified through the NER procedure. We define co-occurrence at the document level, such that two organizations are considered to co-occur if they are mentioned within the same article. We restrict the analysis to the top 100 most frequently mentioned organizations in the corpus to focus on the most salient actors. Pairwise co-occurrences are computed using a counting approach, where edge weights reflect the number of documents in which two organizations are jointly mentioned. To reduce noise from infrequent or incidental associations, we apply a threshold and retain only edges that occur in at least three documents. The resulting network is undirected and weighted, where nodes represent organizations and edges represent the strength of co-mention ties across the corpus.

To identify the internal structure of the network, we use the Louvain algorithm to identify clusters of organizations that frequently co-occur within the same articles, with edge weights corresponding to co-mention frequency. To assess the structural prominence of actors, we calculate degree, weighted degree (strength), betweenness centrality, and eigenvector centrality. Betweenness is computed using the inverse of edge weights so that stronger co-occurrence ties are treated as shorter distances in the network. Eigenvector centrality is used as our primary indicator of network prominence because it captures not only how many ties an actor has, but also whether those ties connect it to other influential actors.

3.2.2 Structural Topic Modeling

In the second step of our analysis, we use Structural Topic Modeling (STM) to identify the main topics that appear in the news' coverage on the intersection between finance and food systems. STM is a helpful tool in that it allows us to infer the latent rather than assume or predefine them. As such, it can be considered as being "unsupervised" (Blei et al., 2003; Roberts et al., 2014). We use STM in order to answer RQ2 and H1 (RQ3) and H2 (RQ4).

STM can exploit both the text of research articles alongside information that is conceptually and empirically connected to the narratives' contents (Jung et al., 2023; Mishra & Jiang, 2022), so it allows us to see how discussions of finance and food systems are related without imposing predefined categories and then build, inductively, a theory that explains these relationships. Furthermore, STM contributes to the granularity of this analysis by revealing the underlying patterns of word usage and associations within the media content. This facilitates a more nuanced understanding of how certain representations of the themes are made and can reveal insights into how media outlets are sites of political and distributive choice.

To do this, our analysis proceeds in four stages. First, the selected keyword corpus underwent pre-processing to optimize it for the STM model. Specifically, we followed a P-N-L-S-W (Punctuation, Numbers, Lowercase, Stopword removal, Word stemming) protocol in the terminology of (Denny & Spirling, 2018). We removed punctuation, numbers and stopwords and converted all words to lowercase, then tokens were stemmed using the Porter algorithm to reduce inflectional forms such as tense and number. Finally, we removed words that appear in fewer than 20 documents to minimize the noise of niche or idiosyncratic words and to improve model convergence. This frequency filtering process removed 44,586 unique terms (and their 112,293 token occurrences) from the corpus. The final corpus thus included 3,189 documents, 4,401 terms and 779,218 tokens.

Next, we determined the number of topics (K)², and we estimate how topic prevalence varies across our three covariates of interest. To do this, after we estimate the topic proportions (θ), we treat these proportions as outcome variables and regress them on document-level covariates supplied in the metadata to estimate the differences in expected topic prevalence conditional on our covariates (in our case, political bias, outlet type, timing of the SDGs, and a smooth term for year of publication). Finally, because θ is estimated and not observed, we compute the confidence intervals by implementing a Bayesian method of composition into the regression stage to account for measurement uncertainty in topic assignments (Roberts et al., 2019). This approach ensures that comparing groups with smaller sample sizes relative to the other (such as finance outlets relative to generalist news outlets) do not lead to artificially precise estimates and instead produces more conservative and statistically robust inferences under sample-size heterogeneity.

Through STM, we uncover not just the 'what' of media framing but also the 'how' and so we can provide insight into the methods of narrative construction, the subtleties of thematic emphasis, and the interplay between media representation and public policy

² Information regarding the selection of K topics as well as diagnostic tests appears in Appendix 1.

discourse. As a result, STM helps bridge the gap between quantitative data and qualitative interpretation, enhancing the rigour and depth of the study's analysis of the world media's discourse on the intersection between finance and sustainable agrifood systems.

4. Results

We begin by showing in Figure 1 how across the 25-year period, media attention to the intersection of capital markets, agrifood systems and sustainability increased. As shown in Panel B (top-right), annual coverage rises from fewer than 50 articles per year in the early 2000s to around 250 articles by the mid-2020s. Growth is particularly pronounced after 2015, given the importance of that year when both the SDGs and the Paris Agreement were signed, and with a brief dip around 2018–2020, before attention increased again.

Coverage is also highly uneven across countries. Panel C (bottom-left) shows that UK outlets account for the largest share of articles in the corpus, followed by France and Australia. In addition, Singapore, Hong Kong, and Malaysia appear as key locations, reflecting their role in Asian financial markets. This suggests that global narratives about the role of capital markets in agrifood sustainability are heavily influenced through media systems in wealthier, finance-intensive economies

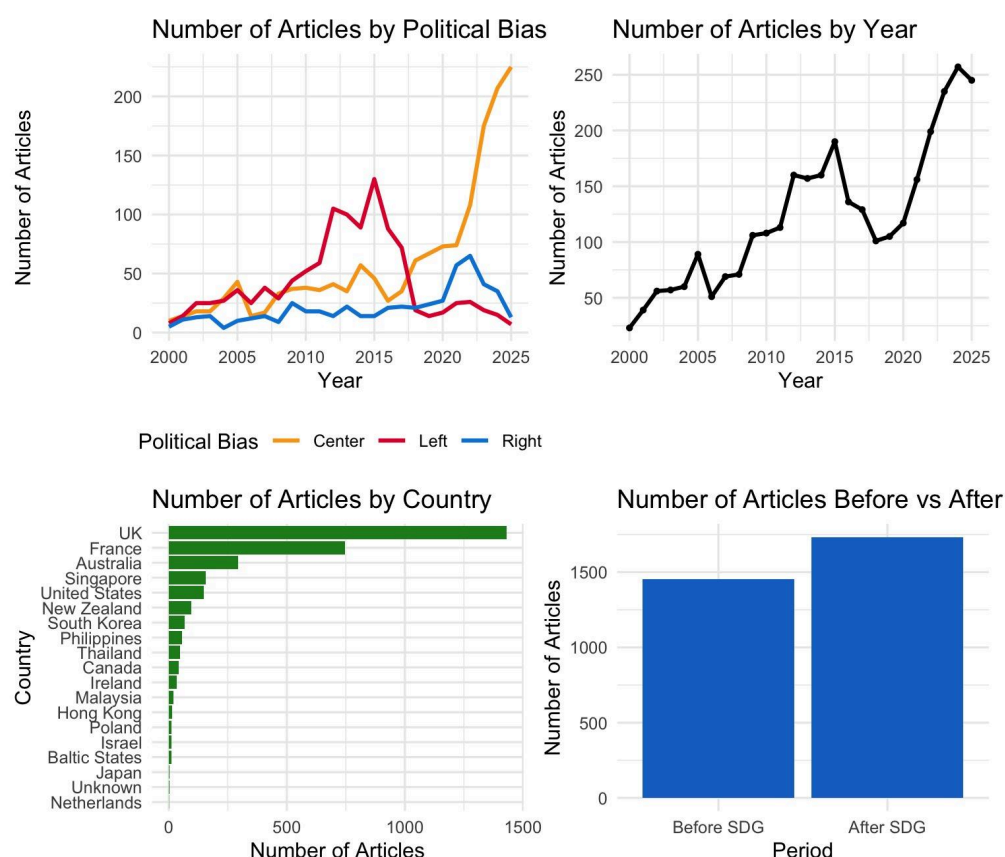


Figure 1. Descriptive Trends

In terms of political orientation, Panel A (top-left) shows that centrist outlets provide the bulk of coverage, particularly after 2020, while left- and right-leaning outlets each contribute smaller but non-trivial volumes, with a particular dominance of left-leaning outlets between 2010 and 2020.

4.1 Dominant organizations

To answer our first research question, we leverage the output of the NER algorithm to identify the organizations more ubiquitous in the discourse in order to create a co-occurrence network to identify the center of gravity. The NER output revealed a dense core composed primarily of international financial and development organisations, such as the World Bank and other development banks, the International Monetary Fund, FAO, the European Union, and the United Nations. These actors are tightly interconnected, often appearing together in coverage of investment initiatives, development programmes, or global summits.

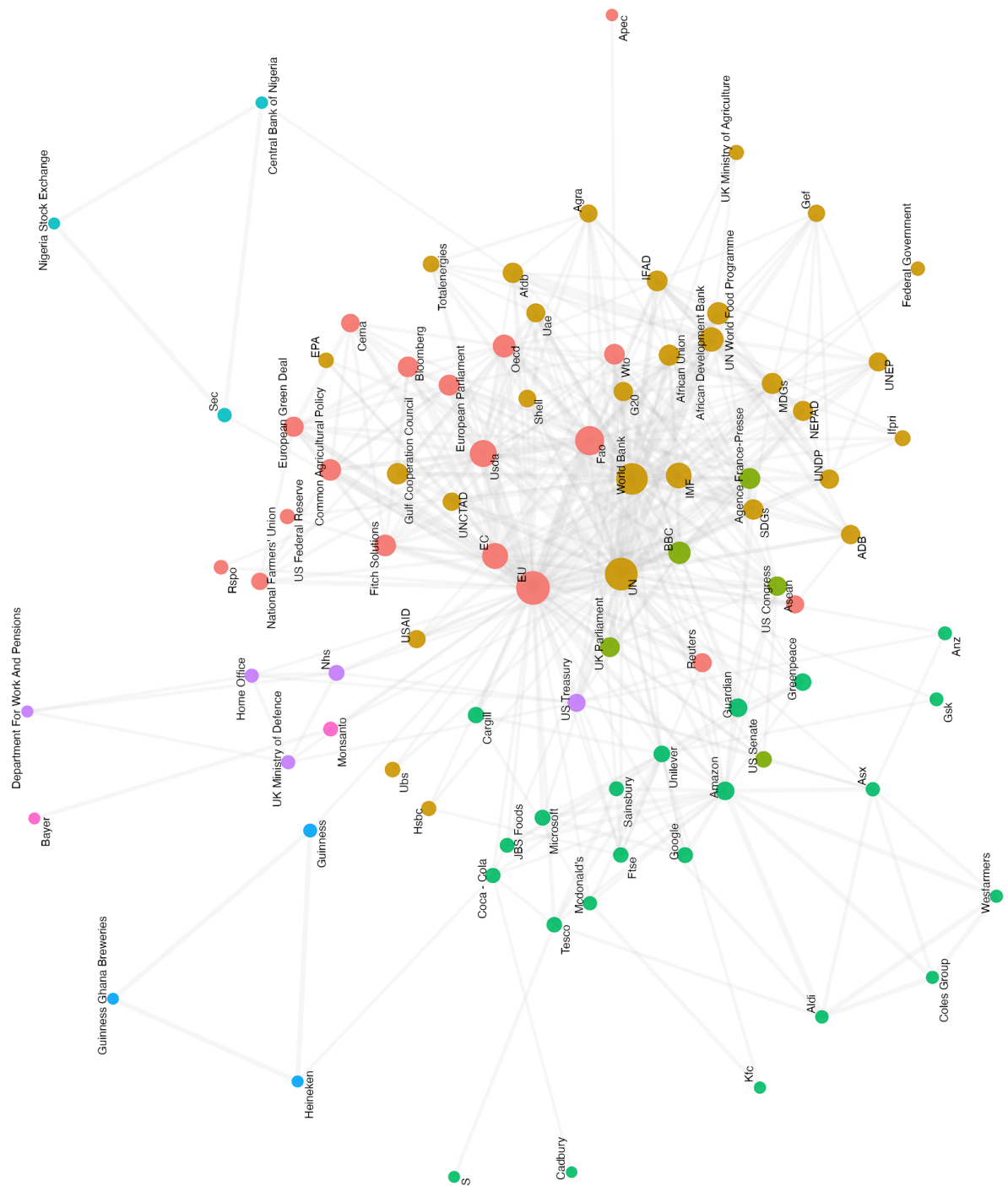
National-level institutions, particularly in the United States and the United Kingdom, also occupy prominent positions. Bodies such as the U.S. Congress, U.S. Treasury, USAID, USDA, the U.K. Parliament, and the U.K. Ministry of Agriculture appear frequently and are deeply embedded within or near this core of the network.

US and UK based National-level ministries of agriculture and rural organisations also widely represented in the network. In contrast, entities representing producers, smallholders or civil society actors appear at the periphery of the network, such as the International Food Policy Research Institute (IFPRI), the Alliance for a Green Revolution in Africa (AGRA), the Roundtable on Sustainable Palm Oil (RSPO), New Partnership for Africa's Development (NEPAD), and the National Farmer's Union with fewer connections and weaker ties. That these organizations have weaker ties to the core is notable given their centrality to on-the-ground agricultural practices.

There is also a wide cluster of multinational food and beverage corporations, supermarket chains, and major consumer brands. The Louvain algorithm identifies this group as a tightly interlinked community separate from the multilateral–development core, indicating that these firms frequently co-occur with one another but their eigenvector centrality is much lower than that of the World Bank, IMF, FAO or UN agencies. This suggests that they do not appear as central agenda-setting actors in the broader finance–sustainability discourse.

This pattern suggests that media coverage of capital markets and agrifood systems is dominated by the perspectives and initiatives of international financial and global governance institutions and large private actors, while the role of farmers and civil society actors is smaller by comparison. We also note that prominent financial institutions (such as the likes of Blackrock or JP Morgan) are also distinctly missing. In terms of agenda-setting, the “centre of gravity” of the discourse is thus clearly located in global financial governance rather than explicitly with financial institutions or local agrarian politics. Taken together, these results indicate that media coverage is anchored in a narrow set of global financial and governance institutions.

Figure 2. Network of Top organizations (sized by eigenvector centrality and colored by Louvain community)



4.2. Thematic structure

Proceeding to our second research question, we examine the structure and evolution of topics in the corpus through structural topic modeling. The STM algorithm identified 30 topics clustered around five distinct thematic clusters: 1) Finance and markets, 2) Food, Agriculture, and Development, 3) Sustainability and natural resources, 4) Trade and Global Supply Chains, and 5) Politics and Governance as shown in Table 2. In addition to this, Figure 3 presents the evolution across time of topic prevalence (*theta*). Several broad patterns emerge.

Table 2. Inferred themes from the corpus

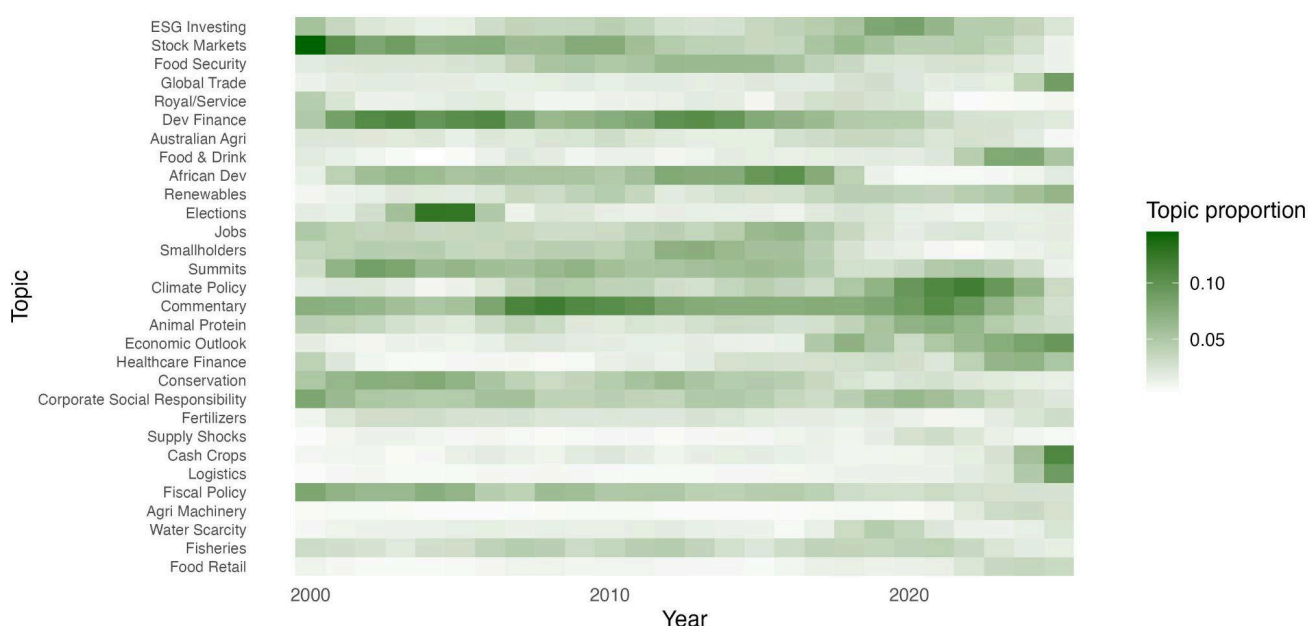
Thematic Cluster	Topic
Finance and Markets	T1 ESG Investing
	T2 Stock Markets
	T18 Economic Outlook
	T19 Healthcare Finance
	T26 Fiscal Policy
Food, Agriculture, and Development	T3 Food Security
	T6 Development Finance
	T7 Australia and agriculture
	T9 African Development
	T13 Smallholders
	T17 Animal Protein
	T22 Fertilizers
	T24 Cash Crops
	T27 Agricultural Machinery
Sustainability and Natural Resources	T10 Renewables
	T15 Climate Policy
	T20 Conservation
	T21 Corporate Social Responsibility
	T28 Water Scarcity
Trade, Infrastructure & Supply Chains	T4 Global Trade
	T8 Food and Drink
	T23 Supply Shocks
	T25 Logistics
	T30 Food Retail
Politics, Governance, and Society	T5 Royalty/Public Service
	T11 Elections
	T12 Jobs
	T14 Summits
	T16 General narrative discourse

Several topics show a clear upward trend, particularly after 2015, such as Climate Policy, Renewables, and ESG Investing. This is consistent with global policy milestones like the Paris Agreement and the Sustainable Development Goals. Other topics show declining prevalence over time. For instance the declining prevalence of smallholders suggests diminishing media attention to small-scale farming and rural livelihoods. The decline of African development and development finance after the global pandemic show a clear shift in priorities. Conversely, the declining prevalence of topics such as fiscal policy and stock market, which can suggest that there is a gradual shift in media attention from traditional financial reporting towards ESG-oriented financial reporting.

We can also see the influence of global events in the distribution of topics. For instance the Millenium Development Goals adopted in 2000 and that were expected to be achieved by 2015 is associated with increased attention to Development Finance and African Development, as well as early peaks in interest in summits reflecting a development-focused global agenda. The transition to the Sustainable Development Goals (SDGs) and the signing of the Paris Agreement in 2015 corresponds to shifts in ESG Investing, renewable energy , and a renewed emphasis on climate policy. This points to the consolidation of sustainability within a financial framework. After 2018, events such as the global pandemic, the war in Ukraine, the cocoa and coffee price surges, and trade conflicts during the Trump presidency produced noticeable spikes in topics related to supply shocks, global trade, cash crops, and logistics, while topics such as, water scarcity and food security received relatively less attention.

Overall, the results of STM results suggest a long-term reframing of agrifood sustainability coverage: from development-oriented concerns in the early 2000s toward financial, climate-aligned narratives in the post-2015 period, punctuated by short-term spikes associated with global crises.

Figure 3. Topic proportion over time



Next we test our hypothesis relating to political bias (H1a and H1b) and issue ownership (H2a and H2b) to address our RQ3 and RQ4 on whether topic prevalence systematically varies according to political orientation and type of outlet.

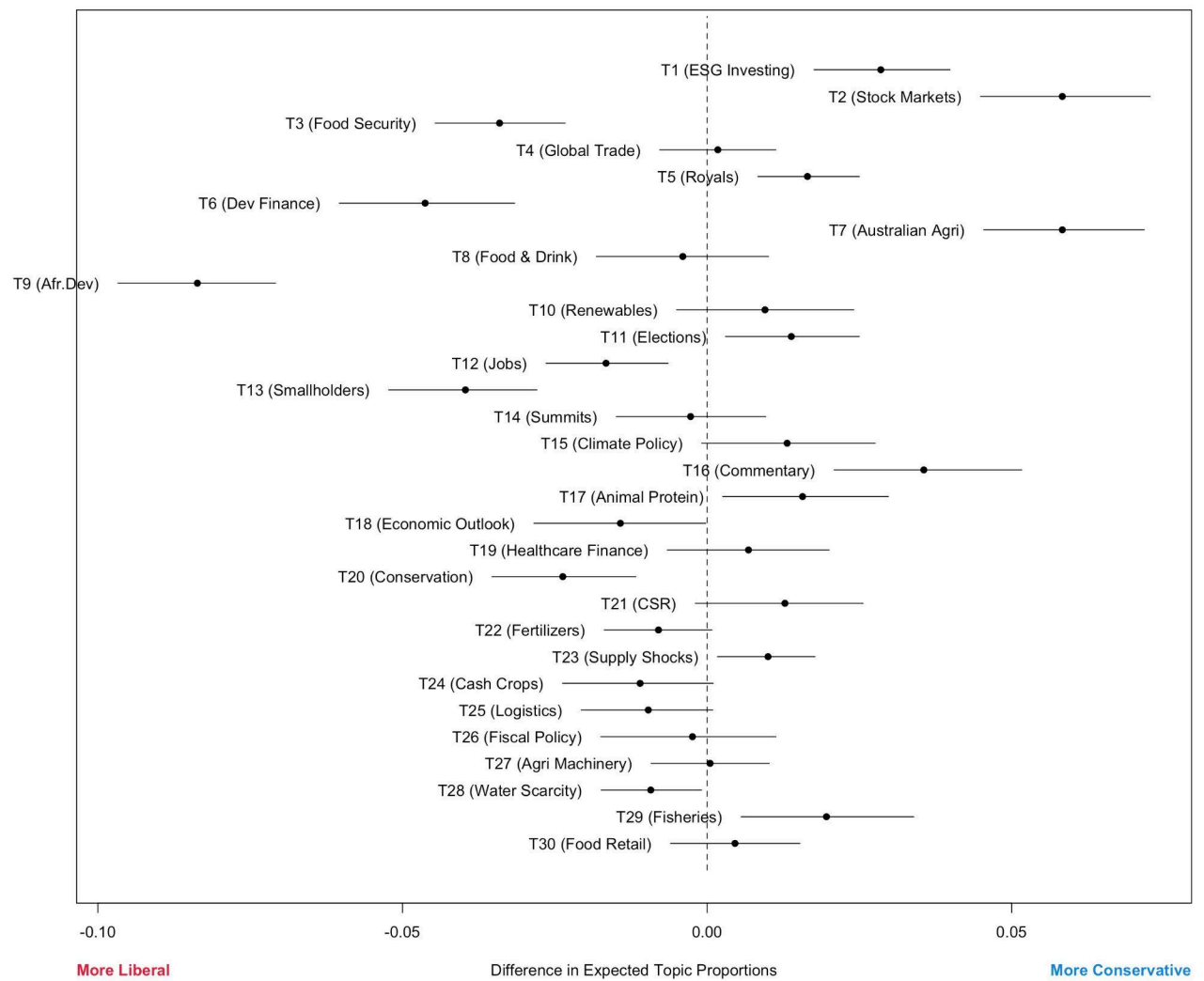
Figure 4 plots the estimated differences in expected topic proportions between right-leaning and left-leaning outlets, holding other covariates constant. Positive values indicate topics more prevalent in conservative outlets; negative values indicate topics more prevalent in liberal outlets. We also include 95% Bayesian credible intervals around each estimate. Because topic proportions are estimated rather than directly observed, these intervals are generated using the method of composition, which carries uncertainty from the STM estimation into the regression stage. The resulting intervals therefore reflect uncertainty in both topic assignment and regression parameters. Estimates whose intervals do not cross zero indicate credibly different topic prevalence across groups.

Consistent with H1a, right-leaning outlets are substantially more likely to emphasise topics that embed a market-liberal understanding of sustainability. Right-leaning coverage devotes more attention to ESG investing (T1) and stock markets (T2) as themes that portray capital markets as central instruments for managing sustainability risks and opportunities. **This is consistent with the logic of ecological modernization** that portrays capital markets as the primary vehicle for greening food systems.

In contrast to conservative outlets, news outlets with left-leaning coverage place greater emphasis on food security (T3), development finance (T6), African development (T9), smallholders (T13), climate policy (T15), and conservation (T20). These topics emphasise inequality, power asymmetries, the livelihoods of producers. This suggests that in left-leaning outlets echoes the concerns that critical scholarship has highlighted on the link between financial and food systems.

Crucially, the results do not suggest that right-leaning outlets ignore sustainability or that left-leaning outlets ignore finance. Rather, they tend to approach and frame sustainability through different lenses. For right-leaning outlets sustainability is embedded within financial instruments and market performance whereas for left-leaning outlets embed the finance within broader debates over justice, vulnerability and state responsibility.

Figure 4. Effect of Political Bias in topic proportion



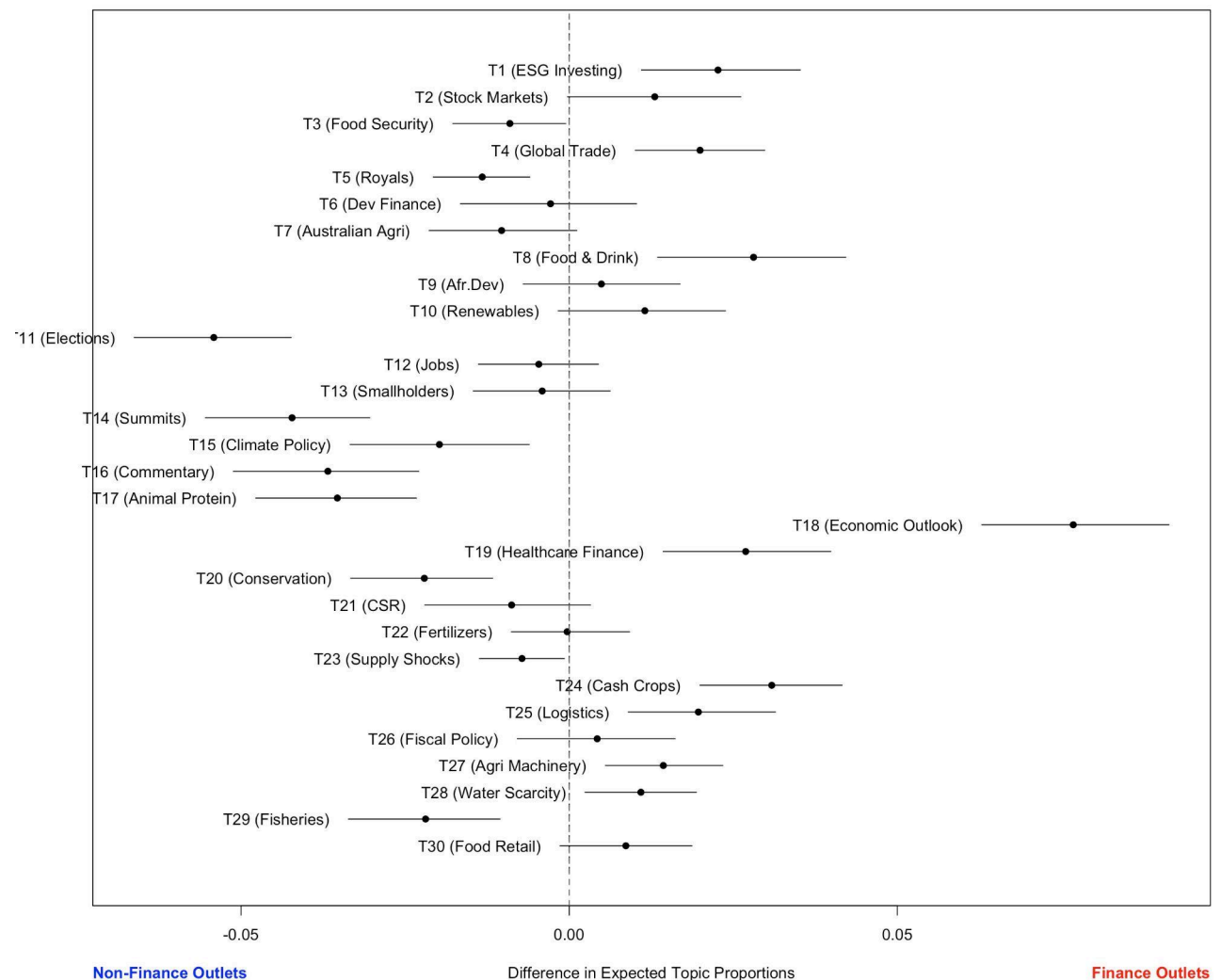
Finally, in Figure 5 we show the estimated differences in expected topic proportions between finance specific news sources and general-interest outlets while controlling for political bias.

The estimates for finance-specialised news outlets strongly support H2a. These outlets display significantly higher expected prevalence for explicitly financial topics, including ESG Investing (T1), Stock Markets (T2), Global Trade (T4), Healthcare Finance (T19), Cash crops (T24) Logistics, (T25) and especially Economic Outlook (T18). Consistent with our theoretical discussion on issue ownership, the professional composition of these news outlets, their expert sources, and audience demands generate incentives to frame sustainability issues primarily through investment risk, market dynamics, and financial strategy.

On the flipside, the evidence does not support H2b. While general-interest outlets are more likely to emphasize governance-adjacent topics, like Food Security (T3), Elections (T11), Summits (T14), and Climate Policy (T15), they rarely reference distributive or

justice-oriented themes our theoretical framework anticipated. Coverage of Development Finance (T6), African Development (T9), Smallholders (T13), and Water Scarcity (T28) does not differ credibly between general-interest and finance-specialized outlets. In headline terms, the topics most closely associated with critical scholarship on the negative impact of finance on food systems are not more prevalent in generalist coverage than in the financial press. This finding is important. The fact that general-interest outlets also do not measurably reorient the discourse toward holistic or justice-oriented framings suggests that a financialised understanding of sustainability has diffused across the entire media ecosystem rather than remaining contained within specialist financial coverage.

Figure 5 Effect of Issue Ownership in topic proportion



Taken together, these results reveal a media landscape in which global financial and global governance institutions are central to the governance of agrifood systems while smaller on-the-ground actors remain in the periphery and where ideological and institutional differences in news sources shape how sustainability is linked to financial markets.

5. Discussion

This paper offers a coherent account of how media discourse at the intersection between capital markets and agrifood systems is neither uniform nor neutral. Our findings shape our understanding of this nexus. We make three claims: that media discourse concentrates attention around a narrow set of financial actors and instruments; that this concentration produces divergent visions of sustainability across the political spectrum; and that the technocentric, market-based framing of agrifood sustainability is now broadly accepted as the norm, even beyond specialist financial media.

Revisiting the research questions and hypotheses. Our first question examines discursive concentration, asking what organizations dominate the conversation (RQ1). The NER analysis revealed that international financial regulators (World Bank, IMF), global governance institutions (FAO, EU bodies, UN agencies), and national-level ministries in high-income countries dominate the discourse. By contrast, civil society organizations, smallholder groups, and producer associations occupy the periphery. These differences are notable given that a sustainable transition in agrifood systems will require multiple levels of governance and often requires more engagement with local knowledge, and bottom-up initiatives. This reflects a form of discursive governance. According to Entman's (1993) theory of framing, the centrality of globally prolific institutions raises questions around what types of solutions may be prioritized and which may be sidelined. By anchoring coverage around actors whose expertise lies in risk management and capital allocation, media narratives constrain sociotechnical imaginaries of sustainable food systems - presenting sustainability as problems primarily of information, pricing, and efficiency; one to be solved by neoliberal and technical means while dismissing alternative perspectives.

Second, with respect to visions of sustainability, we asked, what are the main themes that have emerged in the media covering the role of finance and sustainable agriculture. In this respect, the STM identified 30 distinct topics that were broadly clustered around five thematic groups 1) Capital markets, 2) Food, Agriculture, and Development, 3) Sustainability, 4) Trade and supply chain, and 5) Policy. Across the full corpus, the topics with highest prevalence include those that relate to financial or economic themes such as ESG investing (T1), food security (T3), global trade (T4), and climate policy (T15). Alternatively, development themes such as smallholders (T13) or African development (T9) receive comparatively lower coverage. These results are then corroborated by an inspection of topic proportion over time, which reveals that while there is increased attention to topics related to finance and markets there is also declining attention to smallholders. This indicates a shift toward more market-oriented or technologically driven narratives where sustainability is framed within broader frameworks of financial risk, macroeconomic uncertainty, and ESG transformation issues as consistent with the literature (Clapp, 2019).

Third, we examined how political biases shape narratives and hypothesized that conservative outlets would prioritize market-based topics while liberal outlets would emphasize topics around development and, overall, we find strong support for our hypothesis. Our results illustrate that our understanding of sustainable finance is not neutral

but politically interpreted (Mazzucato & Semieniuk, 2018). As such, that right leaning outlets highlight financial market topics suggests a specific vision of sustainability that aligns with ecological modernization and green growth narratives and, as such, conditions the type of policy solutions that right-leaning outlets prefer. In contrast, while ecological modernization remains central in left-leaning framing as well, it is more so presented as a problem of structural inequality which partially mirrors concerns highlighted by critical scholarship like inequality, justice, and vulnerability. As such, the media does not simply report on sustainable finance debates, it actively constitutes the political window within which these debates unfold, and interventions become feasible or legitimate.

Finally, we examined how discourses varied by outlet and hypothesized that given issue ownership, finance specific outlets would tend to emphasize investment themes while general-interest outlets would emphasize additional themes. In this case, we find that while finance outlets generally emphasize market and investment topics, we found no systematic evidence for H2b. While topics such as Development Finance (T6), African Development (T9), Smallholders (T13), and Water Scarcity (T28) appear in generalist media, their coverage of these topics does not differ systematically from that of finance specific outlets. This suggests that while general-interest media may broaden the discursive context around the role of finance in enabling sustainable agrifood systems, they do not substantively reorient the narrative toward justice-oriented or developmental concerns and consequently, we assert that alternative views of financing sustainable agriculture have failed to break into the mainstream. The fact that general-interest outlets do not substantively reorient the narrative suggests that a financialised understanding of sustainability has become broadly accepted across the media ecosystem.

Taken together, these findings fit within the classic insights from Entman (1993) and McCombs and Shaw (1972). Media coverage does not simply report facts but rather, it actively structures possible and probable futures for sustainable transitions. By foregrounding topics linked to investment risk, disclosure regimes, and market coordination, the media elevates technocratic solutions and consolidates expectations that financial actors are the primary engines of sustainability. This discursive pattern mirrors earlier dynamics observed in the regulation of green finance, where credibility, standardization, and risk management emerged as central organizing principles shaping market behavior. The media's framing therefore functions as a soft form of governance in that it shapes expectations, conditions legitimacy, and influences which policy instruments appear feasible within the political economy of agrifood sustainability.

Equally important, however, is what the STM and NER analyses did not infer from the corpus. Across models, core themes in critical agrifood scholarship such as agroecology, food sovereignty, land grabbing, commodity speculation, structural inequality, failed to appear as coherent topics. Equally, actors that voice alternative approaches and on-the-ground actors were not central in the network. Their absence represents a reporting gap that, in turn, silences the perspectives that challenge incumbent financial and agrifood regimes. That these actors and topics are omitted in the corpus highlights the discursive power of traditional views of sustainability, which ground their legitimacy in the technical

complexity of the policy problems they intend to solve, thereby allowing alternative perspectives to be dismissed as not technically fit for purpose. (Carstensen & Schmidt, 2016). In doing so, these narratives constrain sociotechnical imaginaries of sustainable food systems. Here, the most effective form of power is that which prevents issues from reaching the agenda at all, reflecting the capacity of dominant framing to structure what is thinkable within public discourse. Together, this concentration represents the media's ability to both shape who is heard and what can be said. These findings raise broader questions about the long-term implications of discursive dominance in sustainability governance. If sustainability continues to be defined primarily through financial risk and market efficiency, there is a possibility that these framings will become path-dependent (if they have not already become so), tied to incumbent logics and crowding out alternatives.

These findings also point to a broader dynamic beyond media discourse. Media narratives operate alongside dominant ESG rating methodologies and sustainability reporting frameworks form a wider informational infrastructure through which capital markets interpret sustainability - primarily through the lens of risk. When these information channels converge around a narrow set of themes and actors (as our analysis suggests they do) they may contribute to a form of informational lock-in, in which particular framings of sustainability become self-reinforcing. This suggests that amplifying alternative voices in media coverage around measures that capture dimensions of sustainability beyond financial risk, may be necessary to complement regulatory and financial interventions.

The network structures that emerged from the NER analysis help us identify key leverage points within this ecosystem. A relatively small group of highly central institutions, such as global governance institutions and multinational agrifood firms serve as central nodes that connect separate clusters of actors. As such, their position means that changes in how these actors operationalize sustainability could cascade towards the periphery of the network and reshape not only media narratives but also policy debates and investment practices. As demonstrated in previous studies, interventions targeting central nodes can have system-wide ripple effects (Dordi et al. 2022).

Several limitations should be acknowledged. First, the Nexis Uni database, while extensive, over-represents Anglophone and Global North media systems and may not capture framings prevalent in Global South outlets, where the relationship between finance and food systems may be experienced differently. Second, our headline-based keyword search ensures that articles in the corpus treat the finance-food-sustainability intersection as a primary focus, but it may exclude articles that discuss these themes in the body text. Third, while STM infers latent themes inductively, topic labelling and clustering involve researcher interpretation and alternative configurations are possible. Finally, the covariate analysis identifies associations between outlet characteristics and topic prevalence but cannot establish the direction of causality between editorial supply and audience demand.

6. Conclusion

This study makes both empirical and theoretical contributions to debates on sustainability governance, media framing, and the political economy of agrifood systems. Empirically, it provides the first large-scale, systematic analysis of how global news media construct the relationship between capital markets and sustainable agrifood systems over time. In doing so, the paper sheds light on whose voices and solution pathways dominate media discourse, and how the role of finance in sustainability has become normalized within public debate. Theoretically, we extend the insights from media research on economic policy to the domain of sustainable finance, highlighting how discourse interacts with finance to shape the governance of agrifood system transformations.

Future research could extend this analysis in several directions: examining how media framings translate into investor behaviour and capital allocation decisions, comparing Global North and Global South media ecosystems, and tracking whether specific policy interventions (such as the EU's Sustainable Finance Disclosure Regulation) have altered the discursive landscape. More broadly, this paper suggests that scholars of sustainability transitions should attend not only to the material and institutional dimensions of transformation but also to the discursive infrastructure through which particular transition pathways become thinkable, legitimate, and politically viable.

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